Q1. JEE Main 2026 (23 January Shift 1)

Let the mean and variance of 8 numbers $-10, -7, -1, x, y, 9, 2, 16$ be $\frac{7}{2}$ and $\frac{293}{4}$, respectively. Then the mean of 4 numbers $x, y, x+y+1, |x-y|$ is :

(1) 11 (2) 12 (3) 10 (4) 9

Q2. JEE Main 2026 (24 January Shift 2)

Let $\mathrm{X} = \{x \in \mathrm{N} : 1 \leq x \leq 19\}$ and for some $a, b \in \mathbb{R}$, $\mathrm{Y} = \{ax + b : x \in \mathrm{X}\}$. If the mean and variance of the elements of Y are 30 and 750, respectively, then the sum of all possible values of b is

(1) 60 (2) 100 (3) 80 (4) 20

Q3. JEE Main 2026 (22 January Shift 2)

If the mean deviation about the median of the numbers $\mathrm{k}, 2\mathrm{k}, 3\mathrm{k}, \ldots\ldots, 1000\mathrm{k}$ is 500, then k^2 is equal to :

(1) 9 (2) 1 (3) 4 (4) 16

Q4. JEE Main 2026 (23 January Shift 2)

If the mean and the variance of the data

Class	4-8	8-12	12-16	16-20
Frequency	3	λ	4	7

are μ and 19 respectively, then the value of $\lambda + \mu$ is

(1) 21 (2) 19 (3) 20 (4) 18

Q5. JEE Main 2026 (21 January Shift 1)

Let the mean and variance of 7 observations $2, 4, 10, x, 12, 14, y, x > y$, be 8 and 16 respectively. Two numbers are chosen from $\{1, 2, 3, x-4, y, 5\}$ one after another without replacement, then the probability, that the smaller number among the two chosen numbers is less than 4, is :

(1) $\frac{3}{5}$ (2) $\frac{1}{3}$ (3) $\frac{2}{5}$ (4) $\frac{4}{5}$

Q6. JEE Main 2026 (28 January Shift 1)

The mean and variance of 10 observations are 9 and 34.2, respectively. If 8 of these observations are $2, 3, 5, 10, 11, 13, 15, 21$, then the mean deviation about the median of all the 10 observations is

(1) 4 (2) 7 (3) 5 (4) 6

Q7. JEE Main 2026 (24 January Shift 1)

The mean and variance of a data of 10 observations are 10 and 2, respectively. If an observations α in this data is replaced by β, then the mean and variance become 10.1 and 1.99, respectively. Then $\alpha + \beta$ equals

(1) 15 (2) 5 (3) 10 (4) 20

ANSWERS AND SOLUTIONS

1. (1) **2.** (1) **3.** (3) **4.** (2) **5.** (4) **6.** (3) **7.** (4)

1. **(1)** Mean $= \frac{7}{2}$: $-10-7-1+x+y+9+2+16=28 \Rightarrow x+y=19$.

Variance $= \frac{293}{4}$: $\frac{\sum x_i^2}{8} - \frac{49}{4} = \frac{293}{4} \Rightarrow \sum x_i^2 = 684$.

Known squares: $100+49+1+81+4+256=491$, so $x^2+y^2=193$.

$(x-y)^2 = x^2+y^2-2xy = 193-(361-193) = 25 \Rightarrow |x-y|=5$.

$x+y+1=20$.

Mean of $x, y, x+y+1, |x-y| = \frac{19+20+5}{4} = \frac{44}{4} = 11$.

2. **(1)** Mean of $X = \frac{1+19}{2} = 10$, Variance of $X = \frac{19^2-1}{12} = 30$.

For $Y = aX+b$: Mean of $Y = 10a+b = 30$ and Variance of $Y = a^2 \cdot 30 = 750$.

$a^2 = 25 \Rightarrow a = 5$ or $a = -5$.

If $a=5$: $b = 30-50 = -20$.

If $a=-5$: $b = 30+50 = 80$.

Sum of all possible values of $b = -20+80 = 60$.

3. **(3)** The sequence $k, 2k, 3k, \ldots, 1000k$ has 1000 terms.

The median is $\frac{500k+501k}{2} = \frac{1001k}{2}$.

The mean deviation is $MD = \frac{1}{1000}\sum_{i=1}^{1000} |ik - \text{median}|$.

For $i \le 500$: deviation $= \frac{1001k}{2} - ik$, giving $\sum_{i=1}^{500}(\frac{1001}{2} - i) = 125000$.

By symmetry, the sum for $i \ge 501$ is also 125000.

Thus $MD = \frac{k}{1000} \cdot 250000 = 250k = 500$, so $k=2$ and $k^2=4$.

4. **(2)** Class midpoints: $6, 10, 14, 18$.

Frequencies: $3, \lambda, 4, 7$. $N = 14+\lambda$.

Mean $\mu = \frac{200+10\lambda}{14+\lambda}$.

$\sum f_i x_i^2 = 3(36) + 100\lambda + 4(196) + 7(324) = 3160 + 100\lambda$.

Variance: $19 = \frac{3160+100\lambda}{N} - \mu^2$.

Let $u = 14+\lambda$:

$19u^2 = u(1760+100u) - (60+10u)^2 \Rightarrow 19u^2 - 560u + 3600 = 0$.

$u = \frac{560 \pm 200}{38}$, so $u = 20 \Rightarrow \lambda = 6$.

$\mu = \frac{260}{20} = 13$. Hence $\lambda + \mu = 6+13 = 19$.

5. **(4)** Mean = 8 gives $x+y=14$.

Variance = 16 gives $(x-8)^2 + (y-8)^2 = 4$.

Solving: $x=8$, $y=6$ (since $x>y$).

Set becomes $\{1,2,3,4,6,5\} = \{1,2,3,4,5,6\}$.

Total ways to choose 2 numbers = $\binom{6}{2} = 15$.

Favorable cases (smaller < 4): smaller = 1 gives 5 pairs, smaller = 2 gives 4 pairs, smaller = 3 gives 3 pairs.
Total favorable = 12.
Probability $= \frac{12}{15} = \frac{4}{5}$.

6. **(3)** Given 10 observations with mean = 9 and variance = 34.2.
Eight known values: 2, 3, 5, 10, 11, 13, 15, 21.
Sum of all 10: $\sum x_i = 90$.
Sum of 8 known: 80. So the two unknowns sum to 10.
From variance: $\sum x_i^2 = 10(34.2 + 81) = 1152$.
Sum of squares of 8 known: 1094. So unknown squares sum to 58.
With $x_9 + x_{10} = 10$ and $x_9^2 + x_{10}^2 = 58$: we get $x_9 x_{10} = 21$, giving roots $t^2 - 10t + 21 = 0$, so the unknowns are 3 and 7.
All 10 sorted: 2, 3, 3, 5, 7, 7, 10, 11, 13, 15, 21.
Median = $\frac{7+10}{2} = 8.5$.
Mean deviation = $\frac{1}{10}(6.5 + 5.5 + 5.5 + 3.5 + 1.5 + 1.5 + 2.5 + 4.5 + 6.5 + 12.5) = \frac{50}{10} = 5$

7. **(4)** $n = 10$, mean $= 10$, variance $= 2$.
So $\sum x_i = 100$ and $\sum x_i^2 = 10(2 + 100) = 1020$.
After replacing α by β:
new mean $= 10.1 \Rightarrow \sum x_i - \alpha + \beta = 101 \Rightarrow \beta - \alpha = 1$
New variance $= 1.99 \Rightarrow \frac{\sum x_i^2 - \alpha^2 + \beta^2}{10} - (10.1)^2 = 1.99$
$\sum x_i^2 - \alpha^2 + \beta^2 = 10(1.99 + 102.01) = 1040$
$\beta^2 - \alpha^2 = 1040 - 1020 = 20$
$(\beta - \alpha)(\beta + \alpha) = 20 \Rightarrow 1 \cdot (\alpha + \beta) = 20$
$\alpha + \beta = 20$